

Beyond Dual-Process Theory: Religiosity, Cognition, and Rationalism

Emily Tieu

Department of Cognitive Science, University of Toronto

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Abstract

The current study seeks to examine the role of analytic thinking within dual-process theory and its implications for religiosity, logical reasoning, and the Great Rationality Debate. The dual-process model distinguishes between intuitive and analytic processing, referred to as System 1 (S1) and System 2 (S2) respectively. Empirical evidence presents a consistently negative, albeit weak, correlation between analytic cognitive processing and religiosity, indicating that individuals who are more inclined to engage in S2 processing tend to be more skeptical of supernatural beliefs. This relationship is further scrutinized through the existence of belief bias, where S2 reasoning is thought to promote logical evaluation rather than reliance on prior beliefs. Dual-process theory has faced several critiques, however, as its lack of consideration for individual and cultural differences raises concern about its generalizability. To address these limitations, a new tripartite model is introduced, dividing S2 into a multilevel system. Dual-processing also plays a central role in sparking the rationality wars, where normative rationality of ideal behaviour is contrasted with adaptive, context-sensitive cognition. These perspectives challenge assumptions that religiosity necessarily reflects irrationality. Overall, while dual-processing remains an influential base for explaining human cognition, it must continue to undergo theoretical refinement to account for contextual and individual variability.

Beyond Dual-Process Theory: Religiosity, Cognition, and Rationalism

In recent decades, advances in the cognitive science of religion have focused on explaining supernatural belief through ordinary cognitive processes, one of which being the dichotomy of dual-processing. A crucial postulate of the dual-process model is how human cognition can be dissected into two distinct ways of processing: the first being largely associative and intuitive, and the second classified by its reflective and analytic nature (Pennycook et al., 2013; Tieu, 2026; Vervaeke, Lecture 4, January 30, 2026). These processes are labelled as System 1 (S1) and System 2 (S2) respectively, with theism being predominantly governed by S1 and atheism being predominantly governed by S2 (Tieu, 2026; Vervaeke, Lecture 4, January 30, 2026). Moreover, in lecture 4, Vervaeke (January 30, 2026) describes S1 and S2 to be a continuum of demand on working memory. In accordance with Vervaeke, Evans and Stanovich (2013) affirm the defining attribute of S1 processing to be its autonomy from working memory, whereas S2 processing is best defined by its involvement of working memory. The degree to which each system influences religiosity can therefore be scrutinized by comparing intuitive cognition and its counterpart, analytic thinking (Tieu, 2026; Villanueva, et al., 2024). In this sense, analytic thinking can be regarded as the explicit, sequential processing of rules with respect to S2 (Evans & Stanovich, 2013; Tieu, 2026). Intuitive cognition, by contrast, relies on prior beliefs to make instantaneous decisions, as determined by S1. Therefore, the current study seeks to examine the respective roles of S1 and S2 processing in shaping reasoning performance, particularly in contexts where intuitive beliefs may conflict with analytic evaluation.

An analogous, yet notably distinct, concept is analytic cognitive style, which details the propensity to override highly salient intuitions in problem solving (Pennycook, et al., 2012; Tieu, 2026). It has been evident that analytic cognitive style lowers the acceptance of conventional

religious beliefs, causing an overall decrease in religious engagement (Pennycook, et al., 2012; Tieu, 2026). As such, Pennycook, et al. (2012) deduce that if two people differ in their propensity for thinking, they can obtain different sets of beliefs even if they share the same level of cognitive ability, religious engagement, and ideologies (Tieu, 2026). Importantly, dual-processing theorists distinguish cognitive ability from cognitive style, where cognitive ability is the capacity for analytic thinking while cognitive style is the willingness to perform analytic thinking (Pennycook, et al., 2013; Tieu, 2026). Furthermore, while there has been strong evidence in support of the negative association between analytic thinking and religiosity, it remains unclear what cognitive mechanisms underlying dual-processing take place, and whether a dual-processing model best encapsulates cognitive style. Therefore, understanding the relationship between S1 and S2 processing is crucial for clarifying how differences in cognitive style shape religious commitment, thus begging the question: To what extent does the dual-process model predict religiosity? Analytic thinking only weakly predicts religiosity, though its inherent reflectionism supports better reasoning than intuitive thinking due to its stronger resilience to belief bias; however, the ubiquity of individual differences generates uncertainty regarding the capacity of dual-processing to model problem-solving (Tieu, 2026).

Analytic Thinking and Religiosity

Current evidence and theory suggest that belief in supernatural agents is facilitated by the convergence of intuitive cognitive processes, consistent with S1 processing (Gervais & Norenzayan, 2012). As such, the central argument from a dual-processing account is as follows: if religious belief arises from intuitive processes, and analytic thinking can suppress intuitions relevant to supernatural thinking, then religious belief is predicted to be negatively correlated

with analytic thinking (Baimel et al., 2012). Moreover, it is hypothesized that individuals with a greater tendency toward analytic processing during problem solving are associated with a reduced endorsement of religious beliefs, regardless of cognitive ability (Pennycook et al., 2012). From this perspective, analytic thinkers are believed to be more efficient at overriding conflicts between the material and immaterial world (Pennycook et al., 2014). Pennycook et al. (2014) note that this notion implicitly extends the hypothesis such that intuitive thinkers are less inclined to detect these conflicts, even if a reasoning task does not involve religious belief. Thus, the present paper seeks to examine prior experimental results on individual differences in the propensity to engage S2 processing and their relationship with religiosity.

To assess analytic thinking, the Cognitive Reflection Test (CRT) is often the central metric used, where participants are presented with quasi-mathematical problems with an intuitively compelling, albeit wrong, answer (Baimel et al., 2021; Tieu, 2026; Tosyali & Aktas, 2021). A classic example of what may be asked by participants is: “If it takes 5 machines 5 minutes to make 5 widgets, how many minutes would it take 100 machines to make 100 widgets?” (Baimel et al., 2021), to which it typically elicits the answer of 100 minutes, though the correct answer is 5 minutes. Its simple design permits a reliable way to differentiate associative responses with more reflective responses, revealing more analytic reasoners. Specifically, recognizing that the intuitively compelling answer is incorrect requires the participant to stop and think analytically (Pennycook et al., 2016). As such, CRT performance acts as an index of willingness to employ S2 (Pennycook et al., 2016). In support of the hypothesis, CRT scores were found to be inversely related with religiosity across several meta-analyses (Pennycook et al., 2016; Gervais & Norenzayan, 2012). This correlation appears to be relatively weak ($r = -.18$), though it was consistently negative (Baimel et al., 2021).

Pennycook et al. (2012) deduced this negative correlation to be generally indicative of analytic cognitive styles' effect on religiosity, though measures of cognitive ability were weaker and less consistent. In this light, the tendency to hold empirically vulnerable beliefs may be predicted by the willingness, rather than ability, to think analytically.

Furthermore, Pennycook et al. (2016) ascertain that individuals who identify as atheist scored 18.7% higher than those with religious affiliations, indicating that religious non-believers are evidently more reflective than religious believers. Ståhl and Van Prooijen (2021) corroborate this claim, stating that this empirical support from CRTs can be determined by the motivation for epistemic rationality. That is, an individual's valuation for the extent to which their beliefs are mapped onto the world may stimulate the need to apply cognitive faculties in truth-seeking (Ståhl & Van Prooijen, 2021). However, as discussed in lecture 4, it is discernibly difficult to subjectively map such a large, supernatural world onto the smaller, natural world (Vervaeke, Lecture 4, January 30, 2026). Those who hold epistemic rationality in high regard may be more inclined to engage in analytic thinking when evaluating the plausibility of the existence of deities or other specific religious beliefs (Ståhl & Van Prooijen, 2021). As a consequence, while analytic thinking does have a weak negative association with religious belief, coupling it with other measures such as motivation for epistemic rationality will strengthen this association.

Belief Bias and Logical Reasoning

Over the past few decades, it has been made apparent that people often respond based on believability rather than validity (Pennycook et al., 2013). This incident has since been known as belief bias, and is most evident in syllogistic reasoning tasks, paradigms of logical processing. Participants are provided with a logically valid or invalid conclusion based on two premises,

where the conclusion itself may be believable or unbelievable. Syllogisms therefore probe the cognitive disposition to either bias judgements towards prior belief, or to override prior beliefs with new evidence (Stanovich, 2009). From a dual-processing framework, Pennycook et al. (2013) describe belief biases to arise from an overreliance on intuitive S1 processes to quickly evaluate believability and a failure to engage deliberative S2 processes involved in logical reasoning. Thus, a prominent hypothesis emerged, suggesting that cognitive behavioural biases underlying religiosity may cause a poorer detection for when intuition and logic are in conflict (Daws & Hampshire, 2017). In other words, those who perform better on logic-belief conflict tasks are believed to be more reflective in their problem-solving approaches, thereby spending more time to consider alternative responses (Pennycook et al., 2013).

To directly evaluate the extent to which belief bias affects reasoning, response times (RTs) were used to assess the amount of reflectionism. Taking external factors into consideration, increases in RTs during reasoning may be attributed to S2 processes, as slower responding is typically anticipated by higher levels of deliberation (Pennycook et al., 2013). Accordingly, higher accuracy for conflict problems indicates better avoidance of belief bias. Four combinations of validity and believability are transformed into syllogisms for this test: valid-unbelievable, invalid-believable, valid-believable, and invalid-unbelievable, where the last two present a conflict and the first two do not (Pennycook et al., 2013). As expected, religiosity was strongly negatively correlated with performance on conflict syllogisms, though this was not the case for nonconflict syllogisms (Pennycook et al., 2013). It was also discovered that overall RTs had a positive correlation with accuracy for conflict syllogisms, and that religious belief is negatively correlated with mean RTs for both types of syllogisms (Pennycook et al., 2013). As a result of these studies, it was consistently deduced that such cognitive biases did predict religious

belief, and that they posed a disadvantage in the accuracy of responses to these syllogisms (Daws & Hampshire, 2017; Pennycook et al., 2013; Willard & Norenzayan, 2013). As hypothesized, variation in RTs were found to be caused by differences in the willingness to engage analytic thinking (Pennycook et al., 2013). In terms of dual-processing, mistakes in reasoning are believed to occur when S1 processes are not overridden by S2 processes (Daws & Hampshire, 2017). Critically, the basis of religiosity is not derived from impaired general intelligence, but rather by cognitive-behavioural bias (Daws & Hampshire, 2017). It was thereby confirmed that religious believers had overall poorer reasoning performance, reflecting learnt biases toward intuitive decision making instead of the cognitive ability to maintain working memory (Daws & Hampshire, 2017).

Critiques of Dual-Process Theory

Despite its broad influence, dual-processing has been subject to significant critique, particularly regarding its generalizability, conceptual clarity, and structural assumptions. Evans and Stanovich (2013) define modes of processing to be continuum of cognitive styles involved in S2 processing. Unlike dichotomous systems, modes are known to be sensitive to cultural differences, where analytic styles are observed to vary between Eastern and Western cultures (Evans & Stanovich, 2013). In particular, Eastern cultures are found to be more holistic, where they have a greater tendency to attend to the entire scope of a situation to then assign causality rather than relying on logic (Nisbett et al., 2001). Consequently, this largely contradicts the analytic reasoning observed in and researched by Western cultures. These cultural differences raise concern for the validity of the dual-processing as most research is isolated to Western behaviours, thereby rendering it inadequate as a universal model. Moreover, scholars have

contended that Christianity, the primary religion practiced in the West, maintains strong theological concerns regarding the relations of gods and humans (Nisbett et al., 2001). On the other hand, Eastern religions have historically been deeply integrated into social life, a stark contrast from Western emphasis on secularism (Nisbett et al., 2001). Taken together, these cultural and religious differences suggest that evoked cognitive processes are vastly characterized by sociocultural context, hence challenging the generalizability of dual-process theory as a comprehensive account of reasoning.

To further analyze the efficacy of the dual-processing model to describe reasoning and general fluid intelligence, it is necessary to deconstruct analytic thinking into levels of reflective processing (Evans & Stanovich, 2013; Tieu, 2026). Intelligence itself can be defined as the capacity for picking up on intelligibility that results in knowledge (Tieu, 2026; Vervaeke, Lecture 3, January 23, 2026). Furthermore, intelligence can be split into crystalized intelligence and (general) fluid intelligence; that is, crystalized intelligence refers to the ability to solve problems using knowledge, whereas fluid intelligence is how we use crystalized intelligence (Tieu, 2026; Vervaeke, Lecture 3, January 23, 2026). Thus, it can be inferred that crystalized intelligence is more closely connected to S1 mechanisms while fluid intelligence is more connected to S2 (Tieu, 2026). In consideration of individual differences in cognitive style and intelligence, Stanovich (2010) proposes a new tripartite model of mind, where S1 processing is labelled as the autonomous mind, and S2 processing is divided into two subparts: an algorithmic level and a reflective level. Under this design, the algorithmic mind comprises individual differences in general fluid intelligence, whereas the reflective mind encompasses individual differences in the disposition to engage in rational thinking (Stanovich, 2010). Stanovich and Evans (2013) details the capacity of such thinking dispositions to entail the tendencies to collect information, seek

alternative opinions, and consider future consequences, and other means of extensive thinking before arriving at a conclusion. Therefore, the reflective mind allows insight into individual goals and epistemic values, targeting concerns from the dual-processing model. Ultimately, the dual-processing theory would be better perceived as a multilayer system as opposed to its current dichotomous structure in order to account for all sources of variance.

Implications for the Great Rationality Debate

Beyond its role as a model of cognition, dual-processing raises an overarching question of whether religiosity is irrationality, igniting broader debates about the nature of rationality and the reliability of human thought. Due to the inherently inferential, self-corrective properties of S2, many are keen to believe that increased analytic thinking results in increased rationality (Tieu, 2026). Hence, Vainio (2015) notes that the Great Rationality Debate may be thought of as the extent to which we can trust S1 cognition. Within the realm of cognitive science, one major paradigm of judgement is the normative theory, which concerns the ideal behaviours of perfect rationality (Ganuthula & Dyaram, 2016; Vervaeke, Lecture 4, January 30, 2026). As the central point of discussion in the first phase of the debate, normativity frames heuristic-based responses as deviations from rationality, thereby reinforcing the assumption that S1 is invariably non-normative while S2 is invariably normative (Vainio, 2015; Vervaeke, Lecture 4, January 30, 2026). However, these assumptions are deemed to be fallacies, as there are circumstances where S1 may lead to the correct answers and S2 may be subject to bias (Vainio, 2015). Such errors may be attributed to performance rather than competence, as scrutinized by Vervaeke (January 30, 2026) in lecture 4, where a normative theory of ideal behaviour regarding rationality must be equivalent to a theory of competence. In the extreme, researchers have argued that heuristics and

biases, as associated with S1 processing, do not exhibit any human irrationality (Stanovich, 2011, p. 24). On the other hand, others have asserted that heuristics and biases demonstrate how non-normative responses result not from S2 itself, but rather the intuitive impressions left uncorrected by S2 (Stanovich, 2011, p. 75). In this sense, the rationality wars may be understood as a disagreement extended to the relative roles of intuitive versus analytic processing.

Moreover, adaptive optimization was introduced as a central component of ecological rationalism, offering a key counterpoint in the rationality debates. The goal of ecological rationality is to determine which environmental structures a given heuristic would be successful in (Gigerenzer, 2006, p. 8). Adaptive optimization thereby refers to a strategy for problem-solving, where cognitive processes are shaped to function efficiently within specific environmental constraints (Stanovich, 1999). Notably, Stanovich (1999) emphasizes that rationality in the adaptive sense is not the same as rationality in the normative sense. Thus, logical violations that were previously seen as cognitive fallacies may be reinterpreted as adaptive behaviour (Gigerenzer, 2006, p. 13). By this definition, S1 is more attuned to ecological rationalism the same way S2 is more attuned to normative rationality (Stanovich, 1999). During the second rationality debate, there was an emphasis on finding optimization algorithms depending on the environment (Vervaeke, Lecture 5, February 6, 2026). Furthermore, if S1 cognition is ecologically rational, adapting optimally to specific situations, then religious beliefs may serve functional roles not captured by the normative standards of rationality. This raises the argument that religious belief cannot simply be dismissed as irrational. However, Vervaeke (Lecture 5, February 6, 2026) voiced the issue that there is no *a priori* way of optimizing performance across all possible contexts. Once again, the dual-processing model reveals a limitation in its working-memory definition since there is no surjective mapping onto rationality

(Tieu, 2026; Vervaeke, Lecture 4, January 30, 2026). Accordingly, developments in the Great Rationality Debate have increasingly drawn on models beyond dual-processing theory for a more accurate account of the relationship between religiosity and rationality.

Conclusion

The dual-process theory provides a framework for understanding the relationship between analytic thinking, reasoning performance, and how religiosity relates to cognitive processes. Empirical evidence from conducting CRTs has consistently exhibited a modest negative association between analytic thinking and the propensity to believe and engage in religion. This reduction in religiosity observed among analytic thinkers can be best explained by the inhibition of intuitive cognition rather than an intellectual distinction (Tieu, 2026). Nonetheless, the negative association remains relatively weak unless otherwise coupled with another measure that is more reflective of cultural and individual factors. In correspondence to the individual differences in cognitive style, responsiveness and resistance to belief biases are also vastly influenced by individual differences in working memory (Evans & Stanovich, 2013; Tieu, 2026). While intuitive responses often prioritize coherence with prior beliefs, S2 processing employs working memory to allow individuals to distinguish belief from logic and evaluate arguments based on their structural validity. This capacity supports more reliable reasoning, particularly in situations requiring the detection of conflict between intuitive and logical beliefs.

However, dual-process theory is not without its limitations. Its isolation to Western samples and ideologies present an insufficiency in generalizing the dual-process model as it glosses over both individual differences and the differences in Eastern practices of holistic

processing. Moreover, the interaction between S1 and S2 processes require more dynamical and context-dependent modelling than the traditional dichotomy suggests. These strengths and limitations are hence reflected in the rationality wars, where dual-process theory has both clarified and complicated debates regarding rationality. By framing reasoning in terms of competing cognitive systems, this enabled systematic biases and adaptive judgements to be explicated, though concerns about normativity and how rationality should be assessed were raised. Consequently, although dual-process theory was influential in igniting the beginning phases of the debates, its continued relevance was diminished due to the demand for a more nuanced, integrative account of human cognition.

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